MIL-W-15234B(SHIPS)
16 January 1967
Superseding
MIL-W-15234A
10 May 1951
(See 6.4)

MILITARY SPECIFICATION

WHITING

1. SCOPE

1.1 This specification covers one type of whiting for the manufacture of putty.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

TT-L-215 - Linseed Oil, Raw (for use in Organic Coatings).
TT-P-143 - Paint, Varnish, Lacquer, and Related Materials; Packaging, Packing and Marking of.
TT-W-251 - White-Lead, Basic Carbonate; Dry, Paste-in Oil, and Semi-Paste Containing Volatile Thinner.

STANDARD

FEDERAL

Fed. Std. 141 - Paint, Varnish, Lacquer, and Related Materials; Method of Inspection, Sampling, and Testing.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.- The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

Reagent Chemicals, American Chemical Society Specifications

(Application for copies should be addressed to the American Chemical Society, 1155 Sixteenth, N. W., Washington D. C. 20036).

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Material.- The material shall be in a dry and finely powdered form, and shall be free from deleterious impurities.

3.2 Quantitative requirements.- The whiting shall conform to the quantitative requirements prescribed in Table I when tested in accordance with 4.3.

FSC 8010

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Table I - Quantitative requirements

Characteristics	Requirements	
	Minimum	Maximum
Moisture and other volatile matter, percent by weight	---	2.0
Calcium oxide, percent by weight	50.0	---
Carbon dioxide, percent by weight	40.0	---
Coarse particles retained on No. 325 sieve, percent by weight	---	30.0

3.3 Qualitative requirements. -

3.3.1 Mass color. - When specified (see 6.2), the mass color of the whiting shall acceptably match that of a reference sample agreed upon by the purchaser and the seller (see 4.3).

3.3.2 Sedimentation rate. - When tested as specified in 4.3.3, the opaque suspension shall not settle less than 1.75 inches or more than 3.25 inches in a period of 120 minutes.

3.3.3 Putty properties. - When whiting is mixed with white lead and linseed oil as specified in 4.3.4, the resulting putty shall have the following properties:

3.3.3.1 Working qualities. - When tested as specified in 4.3.4.1, the putty shall show good adhesive and elastic properties and shall not be "short" or "mealy."

3.3.3.2 Ductility. - The putty shall pass the ductility test when tested as specified in 4.3.4.2.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Quality assurance. - Quality assurance shall be provided in accordance with method 1031 of Fed-Std-141.

4.3 Test procedures. - The whiting shall be tested in accordance with the applicable methods and sections specified in Table II. Failure to pass any test shall be cause for rejection of the lot.

Table II - Inspection index

Examination or Test	Test Method		Section of this specification giving requirement
	Applicable method in FED-STD-141	Applicable Section of this specification giving further reference	
Pigment properties:			
Moisture and other volatile matter in pigment	4071	---	Table I
Calcium oxide	7301	4.3.1	Table I
Carbon dioxide	----	4.3.2	Table I
Coarse particles	4101	---	Table I
Mass color	4220	---	3.3.1

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Table II - Inspection index
(Continued)

Examination or Test	Test Method	Test Paragraph	Requirement Paragraph
Pigment Properties:			
Sedimentation rate	---	4.3.3	3.3.2
Putty Properties:			
Working qualities	---	4.3.4.1	3.3.3.1
Ductility	---	4.3.4.2	3.3.3.2

4.3.1 Calcium oxide. - The calcium oxide shall be determined by method 7301 of FED-STD-141 and reported as CaO. To convert percent $CaCO_3$ to percent CaO, multiply percent $CaCO_3$ by 0.560.

4.3.2 Carbon dioxide. - Weigh accurately about 1 gram of the sample into a 300-milliliter Erlenmeyer flask. Add about 25 ml of distilled water and a few drops of phenolphthalein indicator. If the sample (at room temperature) is alkaline to phenolphthalein, titrate against 0.5 normal HCl and record phenolphthalein alkalinity. Add bromcresol green indicator and a measured amount of 0.5 normal HCl to insure excess acidity. Heat until all carbon dioxide is expelled. Cool and titrate the excess acid present against 0.5 normal NaOH. Calculate the difference between the phenolphthalein and bromcresol green end points as CO_2.

1 ml 0.500 N HCl = 0.0110 gm. CO_2

4.3.3 Sedimentation rate 1/. - Dry a portion of the whiting to constant weight at 150°C. Place 9 grams of the dried whiting in a dry glass tube of 12 to 14 mm. inside diameter and 43 to 46 cm. length. Add 45 ml of a dry 1:1 mixture (by volume) of xylene (American Chemical Society reagent grade) and raw linseed oil in accordance with TT-L-215. Cork the tube, shake thoroughly to disperse the whiting, and set it aside for 24 hours. At the end of the 24 hour period, adjust the temperature to 77°F and shake thoroughly to disperse the whiting. Allow the tube to stand quietly in a vertical position at a temperature of 77°F for 120 minutes and measure the distance the opaque suspension has settled out.

4.3.4 Preparation of putty. - Putty shall be prepared by mixing whiting (previously air-dried at 77°F and 50 percent relative humidity) with white lead and raw linseed oil in accordance with the following formula:

Ingredient	Specification	Grams
Whiting	Inspection sample	77
White lead, paste-in-oil (Type B)	TT-W-251	9
Raw linseed oil	TT-L-215	14

4.3.4.1 Working qualities of putty. - Work up the batch of putty to a temperature of 70° F. to 80° F. in the hands, adding additional raw linseed oil, if necessary, to obtain proper consistency. (The additional amount of oil added shall not exceed 3 grams). The resultant putty shall not be sticky. Apply the prepared putty to the edge of a clean piece of window glass and work to a smooth bevel with a putty knife. Spread another portion to a very thin film on the glass. The putty shall show good adhesive and elastic properties and shall not be "short" or "mealy." If the putty does not show these properties, repeat the test preparing fresh materials and allow the thoroughly worked mixture to soak for 48 hours in a nearly filled and tightly sealed container before proceeding.

4.3.4.2 Ductility of putty. - The worked putty (see 4.3.4.1) shall be rolled into a cylindrical-shaped mass, approximately 1 inch in diameter. The cylinder shall be grasped with the hands at opposite ends and slowly pulled. The putty shall have sufficient ductility to permit a reduction in diameter to 1/4 inch or less before rupture.

1/ The test is very sensitive to the presence of moisture. Erratic and nonreproducible results will be obtained if the sample of whiting, the reagents (xylene and linseed oil), the sample tube, and the cork stopper are not completely dry. In the event that results are obtained that are not reproducible, the entire test should be repeated with completely dry materials. The inside diameter of the sample tube is also quite important and tubes that do not exactly conform to the requirements should not be used.

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5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Unless otherwise specified, the whiting shall be packaged level C in accordance with TT-P-143. The unit containers shall be of the size specified (see 6.2). Preparation for delivery shall conform to TT-P-143, except that marking of the following information shall be by lithographic process or stencil:

(a) Name of material.
(b) Stock No.
(c) MIL-Spec. MIL-W-15234B.
(d) Name of manufacturer.
(e) Manufacturer's batch number.
(f) Date of manufacture.
(g) Government contract and/or number.

5.2 Packing. - Unless otherwise specified, the whiting shall be packed level C in accordance with TT-P-143.

6. NOTE

6.1 Intended use. - Whiting, as described in the specification, is intended for use as an ingredient in putty. It is essentially pure calcium carbonate. It is a bulky pigment, quite stable to light, and is unaffected by hydrogen sulfide. It is practically insoluble in water, but is soluble in all dilute acids with the evolution of carbon dioxide. A precipitated grade of calcium carbonate for use as a paint ingredient is covered by MIL-C-15198.

6.2 Ordering data. - Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) A color standard where required (see 3.3.1).
(c) Size of unit container (see 5.1.1).

6.3 Whiting should be purchased by net weight.

6.4 CHANGES FROM PREVIOUS ISSUE. THE EXTENT OF CHANGES (DELETIONS, ADDITIONS, ETC.) PRECLUDE THE ANNOTATION OF THE INDIVIDUAL CHANGES FROM THE PREVIOUS ISSUE OF THIS DOCUMENT.

Preparing activity:
Navy - SH
(Project 8010-N063Sh)

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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